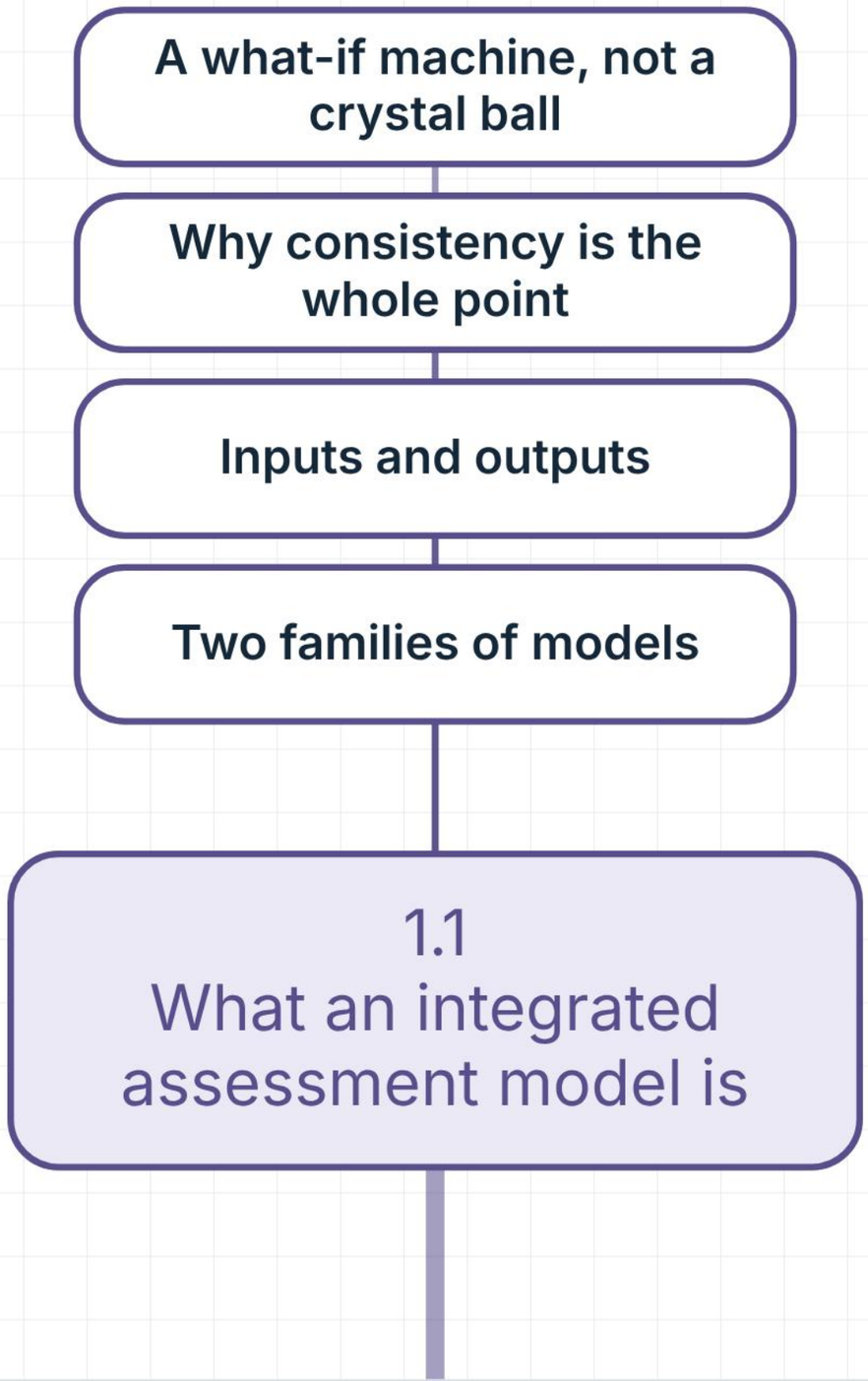


Go 1: See the whole picture

Understand what GCAM and other integrated assessment models are for, what goes in, what comes out, and how to read a scenario.

6 modules. Press Next to zoom into each one.



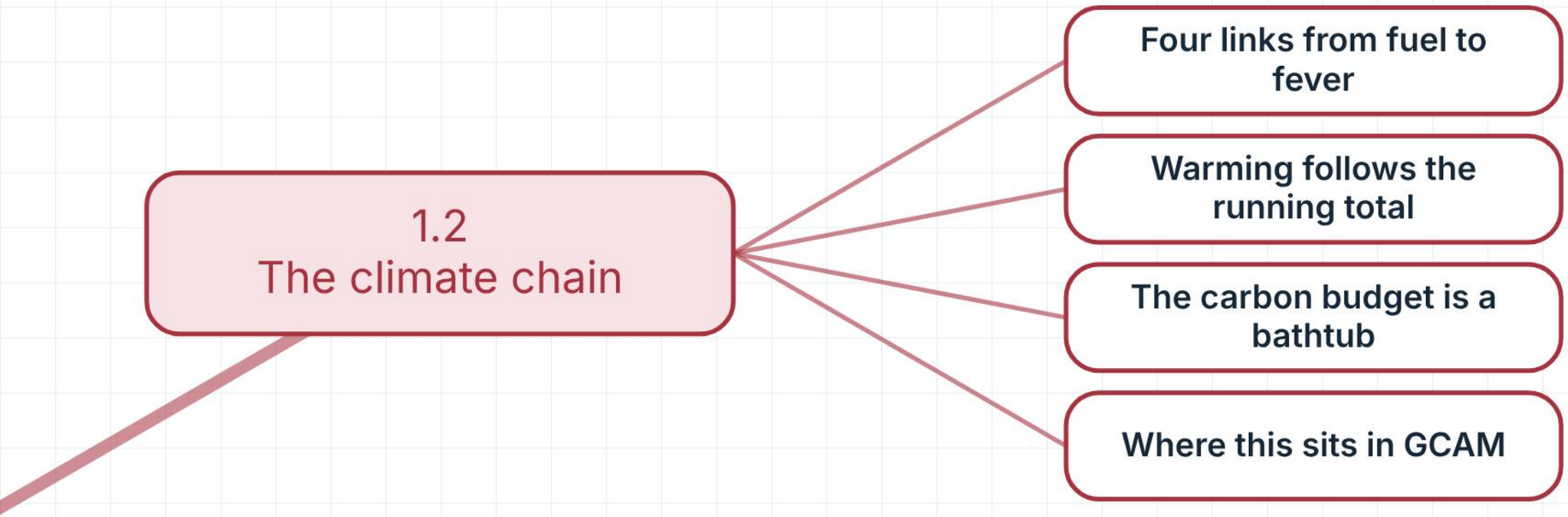
1.1 What an integrated assessment model is

Why a model of the year 2100 is a consistent what-if machine, not a crystal ball, and what goes in and comes out of one.

Remember

- An IAM is a consistent what-if machine, not a forecast.
- Inputs are assumptions you choose, including policies; outputs are projections.
- Consistency (energy, emissions and warming all agree) is what an IAM adds.
- Cost-benefit models ask how much to cut; process-based models like GCAM ask how a goal can be met.

Open module 1.1



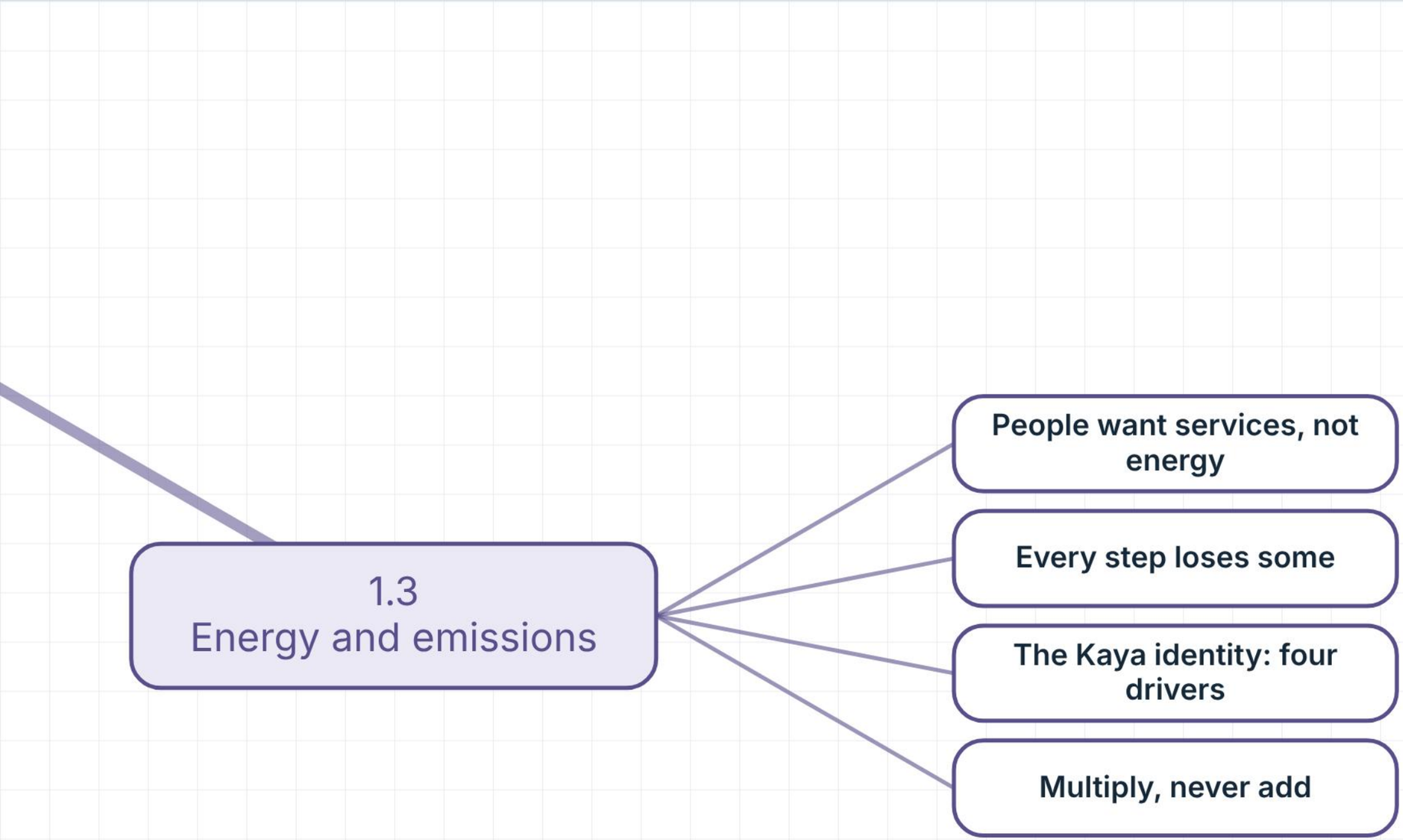
1.2 The climate chain

From emissions to warming in four links, why warming follows the running total of CO2, and the carbon budget as a bathtub.

Remember

- Emissions → concentration → forcing → temperature: four links in every IAM.
- CO2 stays for centuries, so warming follows the running total of emissions.
- Rule of thumb: about 0.45 °C per 1,000 Gt of CO2 (likely 0.27–0.63).
- About 500 Gt was left from 2020 for 1.5 °C (50% chance): 12.5 years at 40 Gt a year.

Open module 1.2



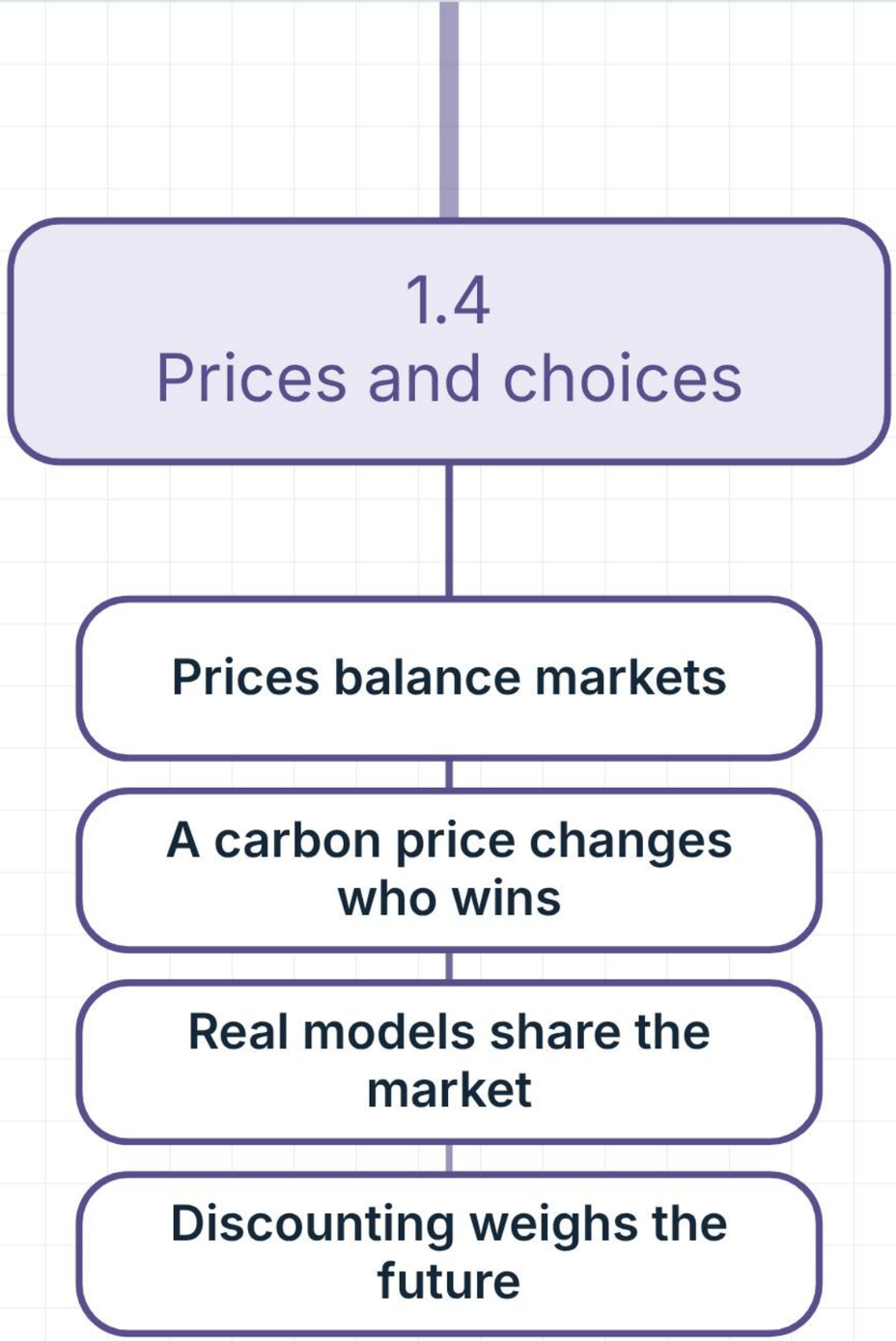
1.3 Energy and emissions

Why people want energy services, not energy, how much is lost along the chain, and the Kaya identity: four drivers that multiply into CO2.

Remember

- People want energy services; energy flows from primary to final energy to services.
- Every conversion step loses energy; in the example only 36 of 100 units reach homes.
- Kaya: CO2 = people × income per person × energy per $ × CO2 per energy.
- Society sets the first two factors; technology sets the last two.

Open module 1.3



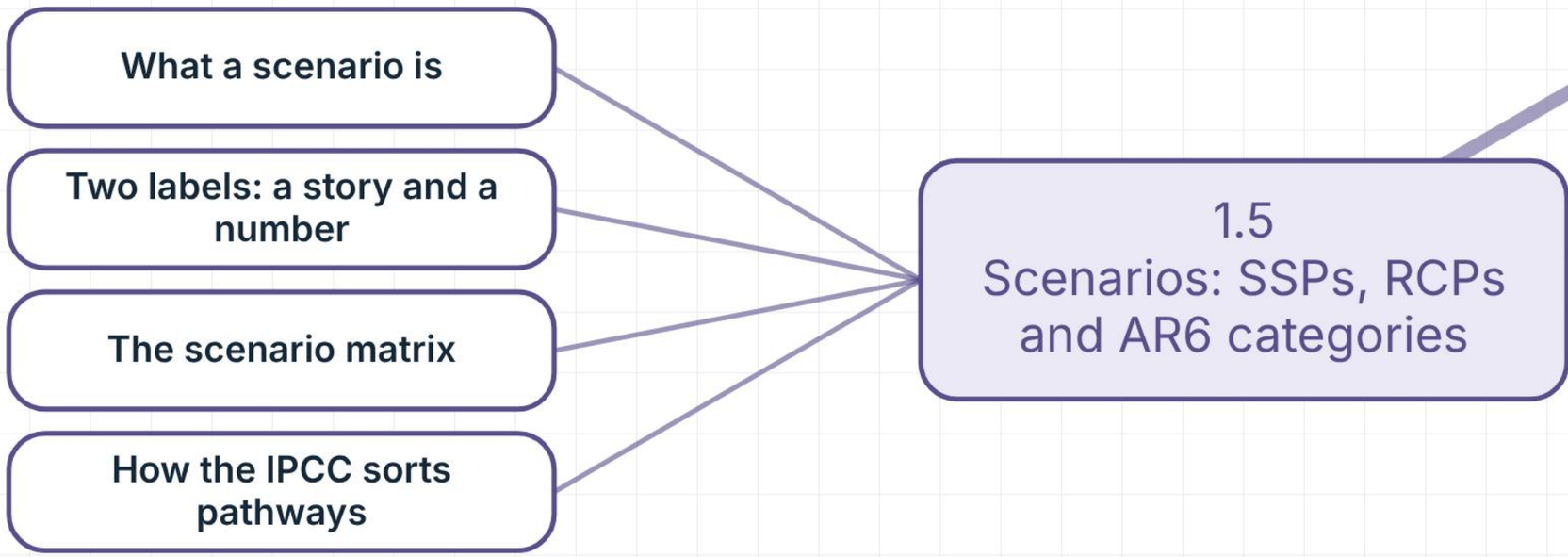
1.4 Prices and choices

Three economic ideas behind every IAM: prices balance markets, a carbon price changes who wins, and discounting weighs the future.

Remember

- Prices move until supply equals demand: the equilibrium.
- A carbon price adds emission factor × price to each option's cost, so clean options gain.
- Toy break-evens: gas beats coal above $20/t, solar beats gas above $25/t.
- Real IAMs share markets by relative cost instead of picking one winner.

Open module 1.4



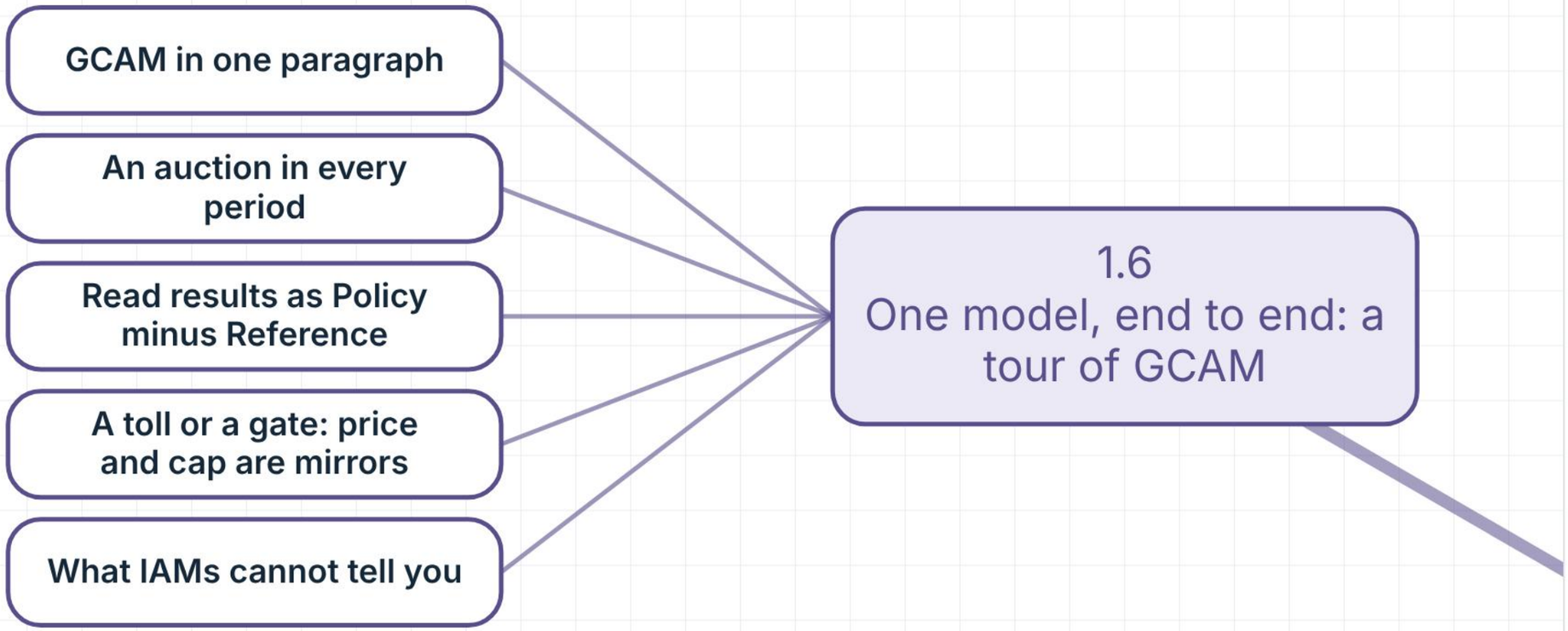
1.5 Scenarios: SSPs, RCPs and AR6 categories

How the research community builds futures: five stories about society (SSPs), forcing levels (RCPs), the matrix that joins them, and how the IPCC sorts pathways.

Remember

- A scenario is a consistent what-if story with numbers, not a forecast or a probability.
- SSP = story about society; the number = forcing in 2100 in W/m², not °C.
- Some SSP × forcing cells cannot be reached, for example SSP3 at the lowest levels.
- The IPCC's 2022 report sorted pathways into categories C1 (1.5 °C) to C8 (above 4 °C).

Open module 1.5



1.6 One model, end to end: a tour of GCAM

A guided tour of GCAM: 32 regions, markets cleared every five years, linked energy, land and water, Hector inside; then reference versus policy runs and what IAMs cannot tell us.

Remember

- GCAM: 32 regions, 300+ land units, 235 basins; 2021 base year, then 5-year steps to 2100.
- Each period, prices adjust until every market clears; there is no foresight.
- Energy, land and water are linked; emissions go to Hector, which returns temperature.
- Read results as Policy minus Reference; a reference is a yardstick, not a forecast.

Open module 1.6